

Test Cases:

Valid Test Cases

#	Input	Level	Expected Result	Actual Result
1	x	Level 1	VALID identifier	VALID identifier
2	total1	Level 1	VALID identifier	VALID identifier
3	sum2	Level 1	VALID identifier	VALID identifier
4	_var	Level 1	VALID identifier	VALID identifier
5	begin int x; x = 10; print(x); end	Level 2 & 3	All tokens found; parse SUCCESS	Tokens identified correctly; Program syntactically CORRECT
6	begin int y; y = x + 5; print(y); end	Level 3	Parse SUCCESS	Program syntactically CORRECT

Invalid Test Cases

#	Input	Level	Expected Result	Actual Result
1	1x	Level 1	INVALID — starts with digit	INVALID identifier
2	ab-c	Level 1	INVALID — contains hyphen	INVALID identifier
3	a\$b	Level 1	INVALID — contains \$	INVALID identifier
4	(empty string)	Level 1	INVALID — empty input	INVALID identifier
5	begin int x end	Level 3	FAILED — missing ;	Syntax ERROR detected; parse FAILED
6	begin int 1x; end	Level 2	Lex error: 1x invalid	1 recognized as NUMBER and x as IDENTIFIER; syntax/lexical error generated